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Abstract

In this presentation, we created colorized images of different messier objects (Crab Nebula, Orion Nebula, and the Butterfly Galaxies) in space. Our overarching goal was to learn more about space than the prior knowledge we entered the class with. To create said images we used pictures from the school's telescope along with the editing software of both GIMP and MaxIm DL. The result was colored pictures, and a further exposed piece of astronomy. The overall significance was the newfound knowledge of different objects in space.

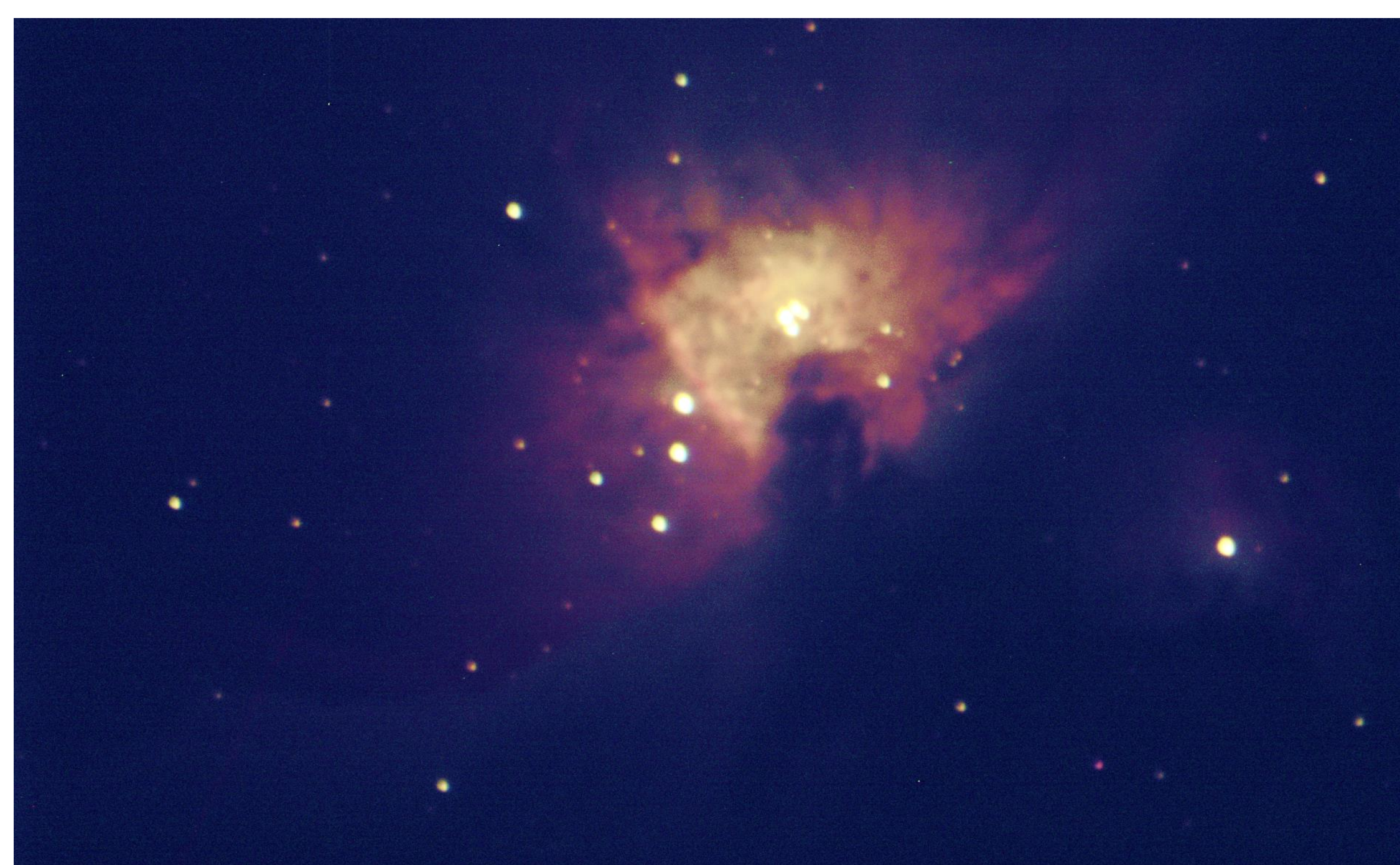


The Hubble and James Webb telescope (left) are telescopes owned by NASA. Many of the photos you see online are from these telescopes, however we only used pictures taken from the BSU telescope (above).

Methods

- Gather photos from the Bridgewater State University's telescope
- Calibrate the photos from the telescope using Maxim DL. This is done by using flat and dark images to calibrate our images.
- Export the stacked files to GIMP
- Use GIMP to edit the photos to colorize and brighten the picture using the curves tool.
- Eliminate blur by using filters on GIMP.

Results

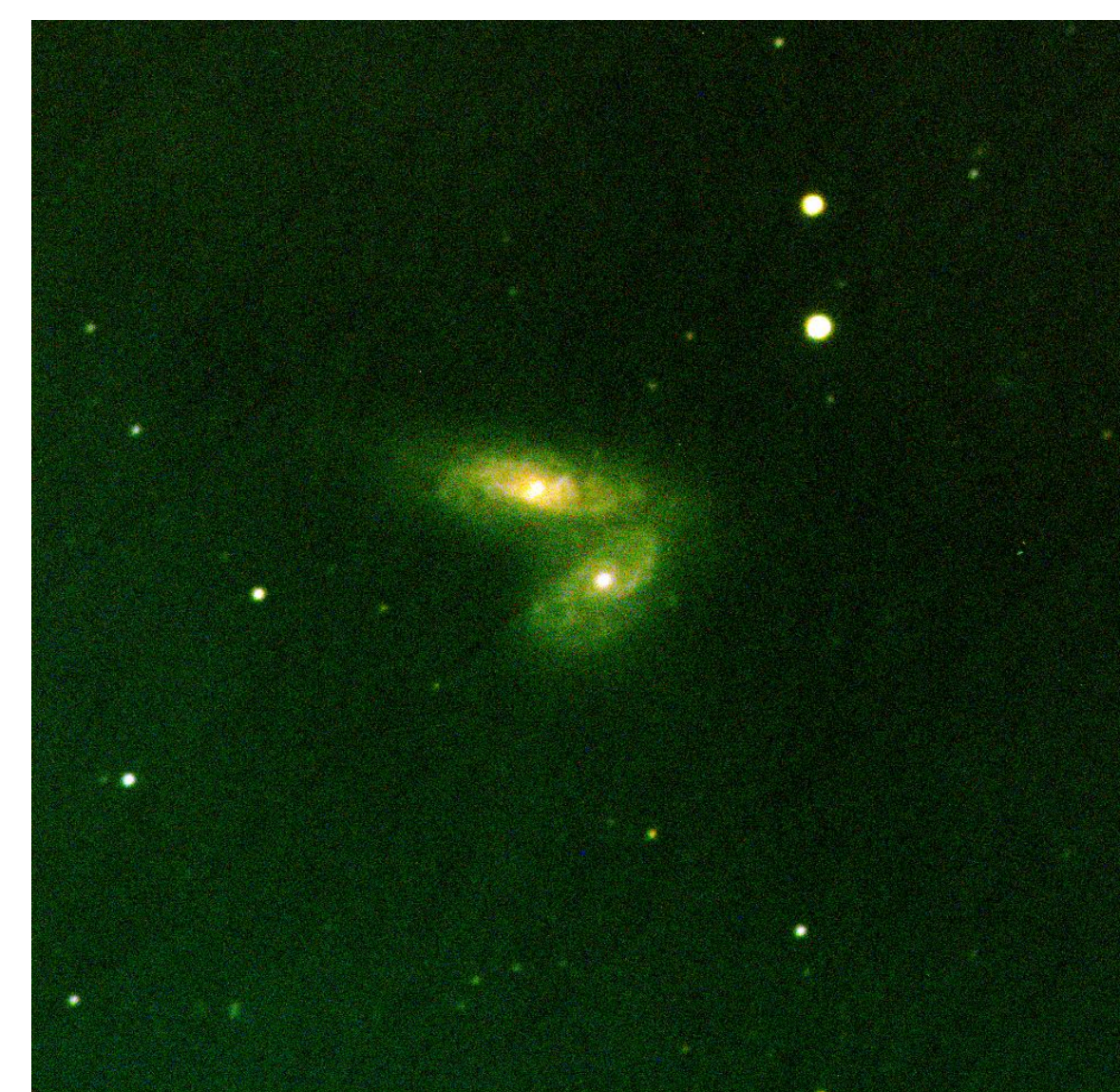


Orion Nebula (Messier 42)

Located 1,344 light years in the Milky Way galaxy. It is a diffuse nebula south of the Orion belt in the Orion constellation Known as the middle star in the Sword of Orion.

Butterfly Galaxies (Siamese Twins/NGC 4567 and NGC 4568)

Located in the Virgo Cluster of Galaxies. These two galaxies are in the process of merging and will conjoin in about 500 million years.



Crab Nebula (Messier 1)

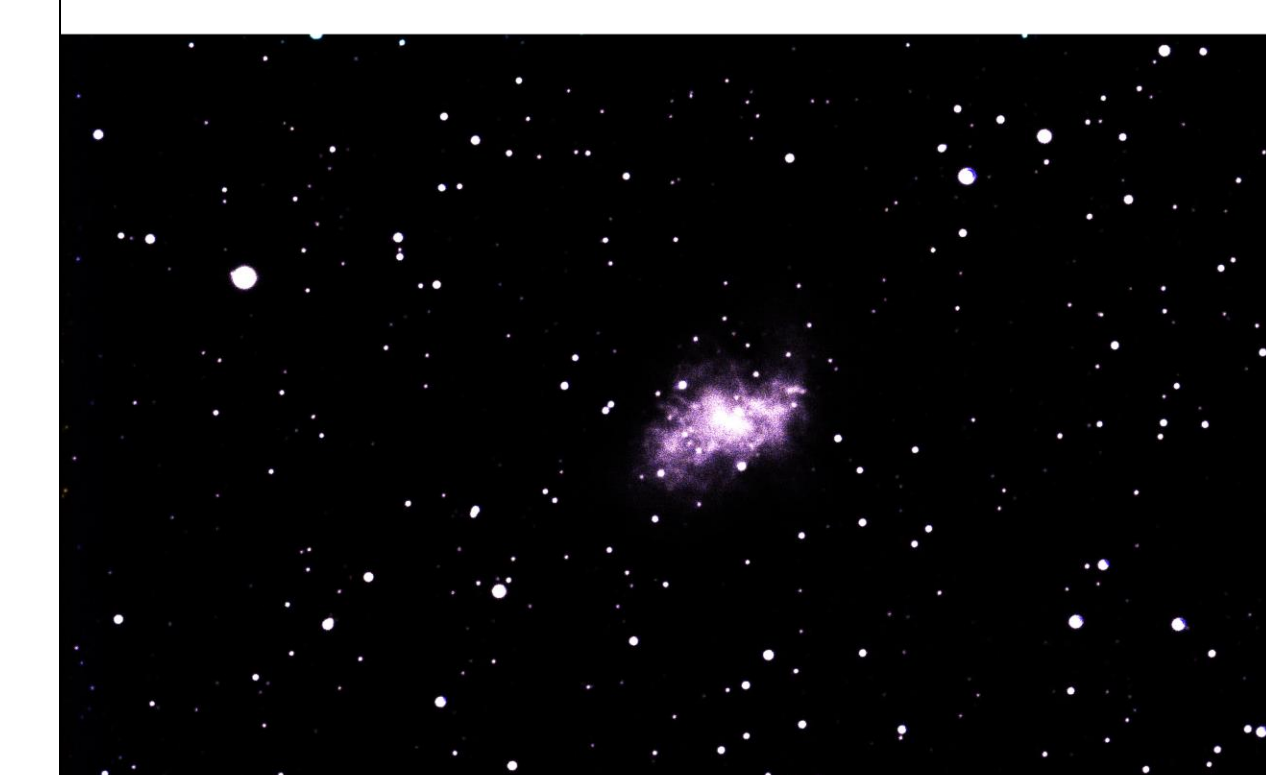
Located about 6500 light years in the Taurus constellation, the crab nebula is the result of supernova explosion that occurred only about 1000 years ago. Since, it has only expanded further and spans about 11 light years wide.



Discussion

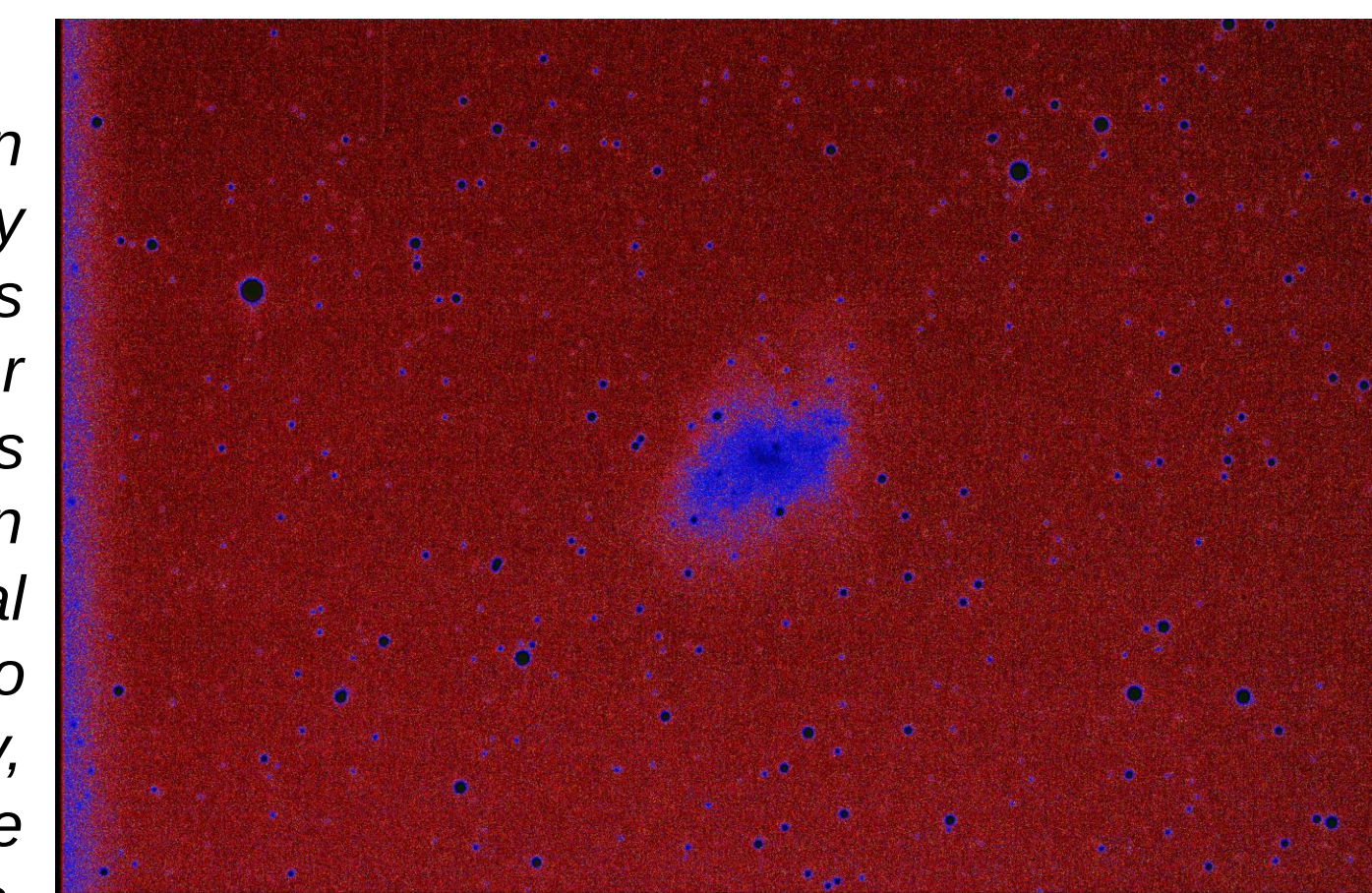
Andrew Sanft:

To the left is a starting picture I made of the Crab Nebula. The drastic difference from beginning to end is quite obvious. My experiences taught me more about not only space but perseverance in a topic I had little background knowledge of. I also improved in my overall ability in GIMP and the such tools. With the new knowledge, I was able to make a more realistic Orion nebula picture



Thomas Berek

On the right is my first attempt on editing the Crab Nebula. I clearly did not know what I was doing as it looks like a portal to another dimension. This was photo was edited with GIMP, and since then I have improved given the final results. I have been able to control the colors more effectively, and make the pictures look more realistic.



Acknowledgements

Thank you to the BSU observatory and my mentor Jamie Kern. Without their help this presentation and our results would not be possible.

Works Cited

“Messier 1 (The Crab Nebula).” *NASA*, NASA, science.nasa.gov/mission/hubble/science/explore-the-night-sky/hubble-messier-catalog/messier-1/. Accessed 17 Apr. 2024.

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